



AI & ML

Group A

Proposed Dataset

GROUP MEMBERS

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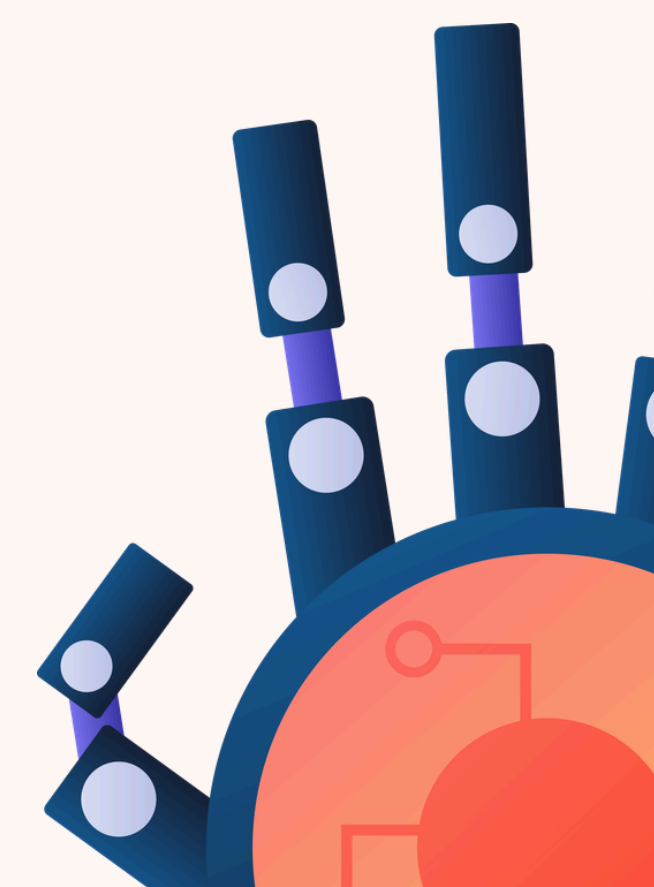
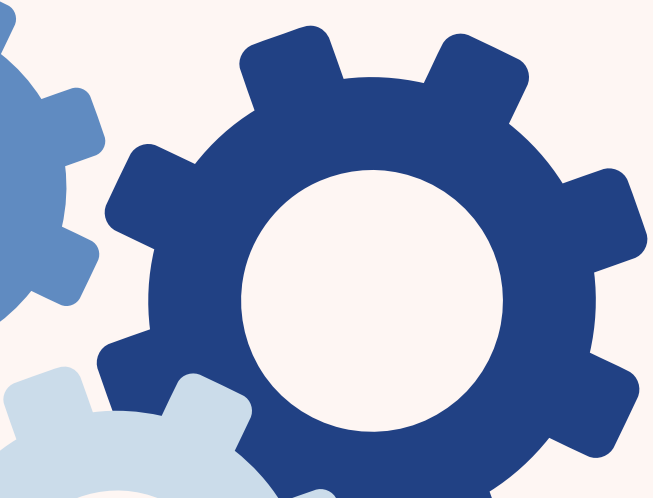
INTRODUCTION

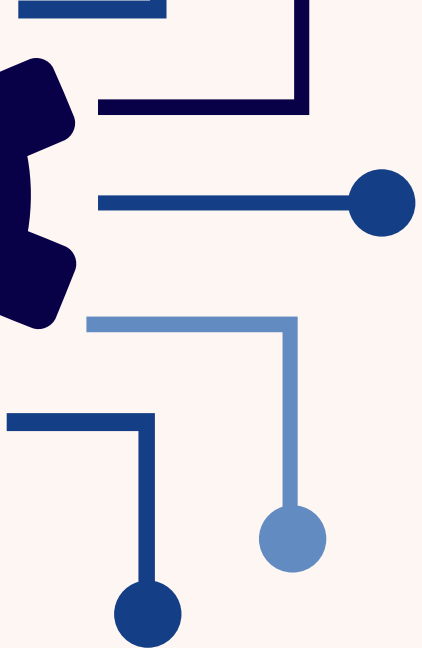
Research Question

Can a Convolutional Neural Network accurately distinguish between 15 specific fruit varieties using a controlled subset of the Fruits-360 dataset?

Title

A Performance Analysis of Deep Learning Architectures for the Classification of 15 Selected Fruit Varieties using CNNs and Transfer Learning





DATASET

Source

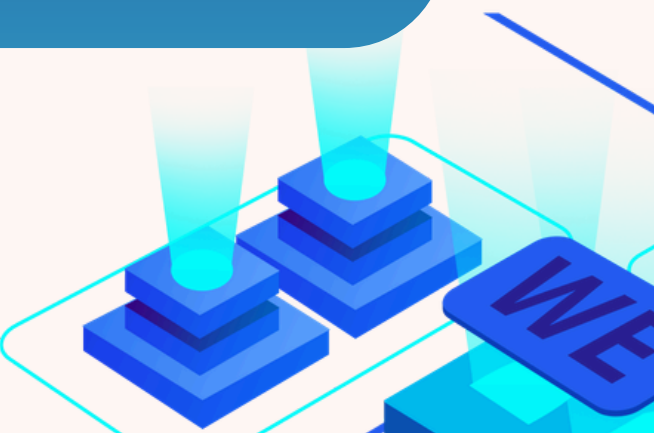
[Fruits-360 dataset \(Kaggle\)](#)

Original Dataset Composition

Total images: 180079
Training size: 135071
Test set size: 45008
Number of classes: 257

Decomposed Dataset Composition

Total images: 10192
Training size: 7635
Test set size: 2557
Number of classes: 15



ANALYSIS

Extraction Method: We manually filtered the larger fruits 360 library to extract only 15 categories.

Labeling Approach: The dataset uses automated labeling based on directory structures. Each image is stored in a folder named after its fruit type, which serves as the ground truth label for the classification algorithm.

Image Acquisition: The original authors captured these images by placing fruits on a rotating motor at 3rpm and filming them for 20 seconds under consistent lighting against a white background.

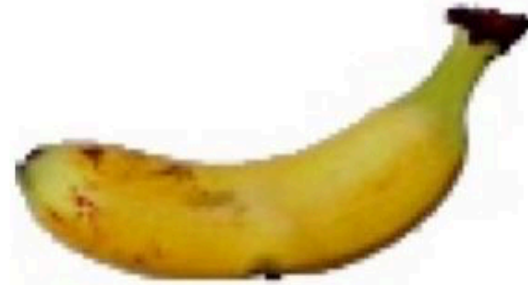
SAMPLE VISUALIZATION

Sample Images from Selected 15 Fruit Classes

Apple 8



Banana Lady Finger 1



Blueberry 1



Carambola 1



Dates 1



Fig 1



Guava 1



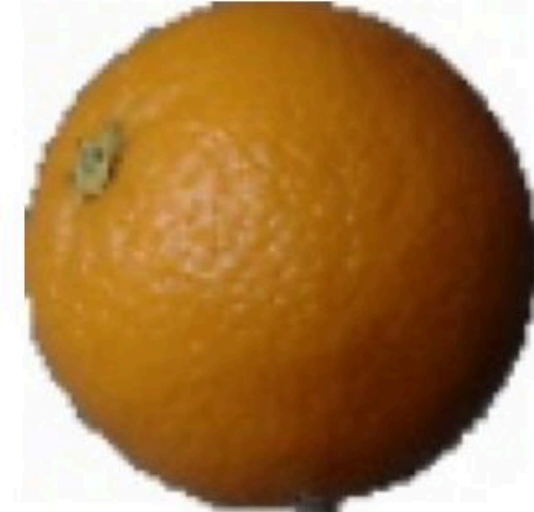
Kiwi 1



Mango 1



Orange 1



Pepper 2



Pineapple 1



Rambutan 1



Strawberry 1



Watermelon 1



SELECTED CLASSES

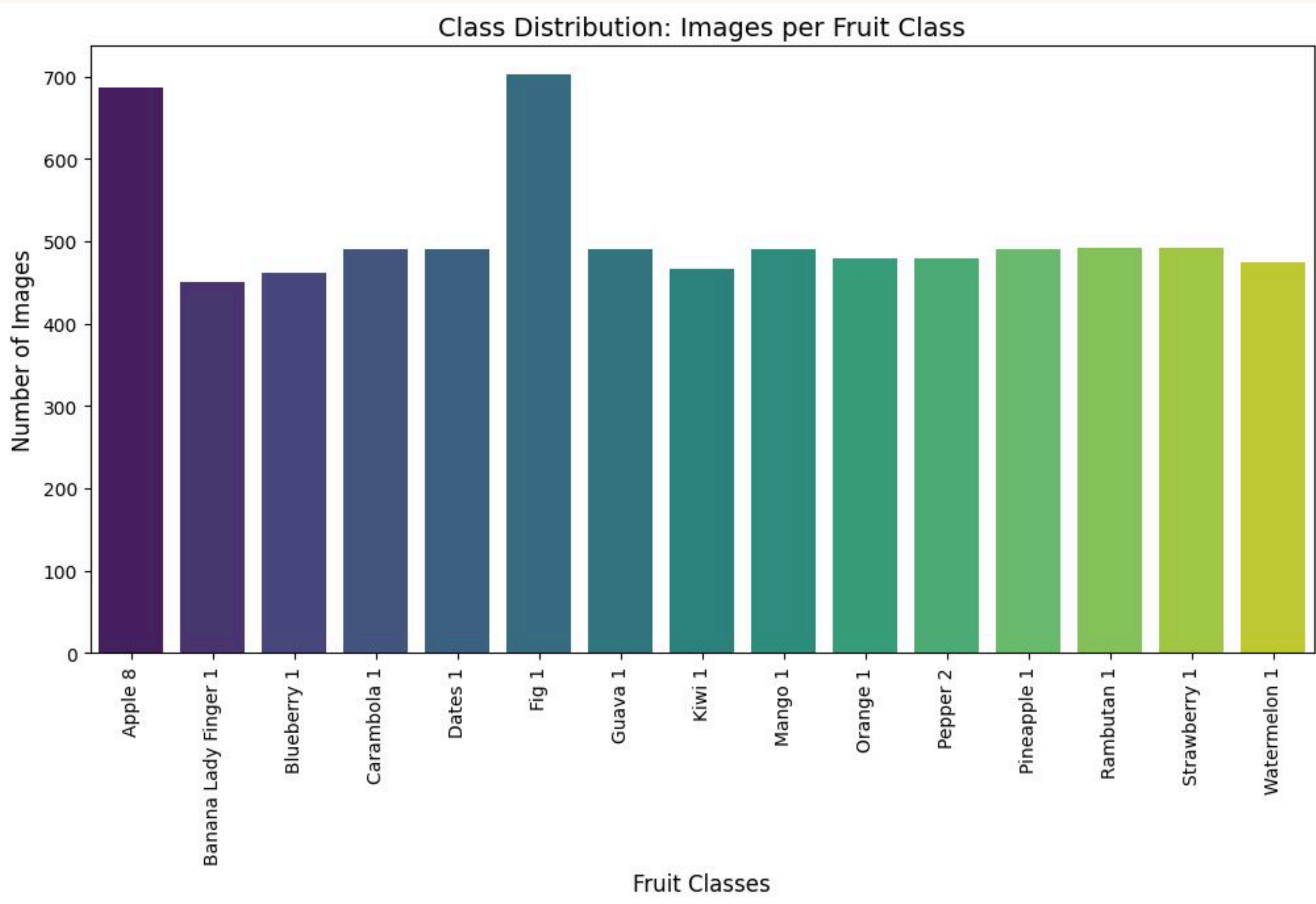
Classes (Kaggle dir name)	Training	Test
Apple (Apple 8)	687	228
Banana (Banana Lady Finger 1)	450	152
Blueberry (Blueberry 1)	462	154
Carambola (Carambola 1)	490	166
Dates (Dates 1)	490	166
Fig (Fig 1)	702	234
Guava (Guava 1)	490	166
Kiwi (Kiwi 1)	466	156

Classes (Kaggle dir name)	Training	Test
Mango (Mango 1)	490	166
Orange (Orange 1)	479	160
Pepper (Pepper 2)	480	158
Pineapple (Pineapple 1)	490	166
Rambutan (Rambutan 1)	492	164
Strawberry (Strawberry 1)	492	164
Watermelon (Watermelon 1)	475	157

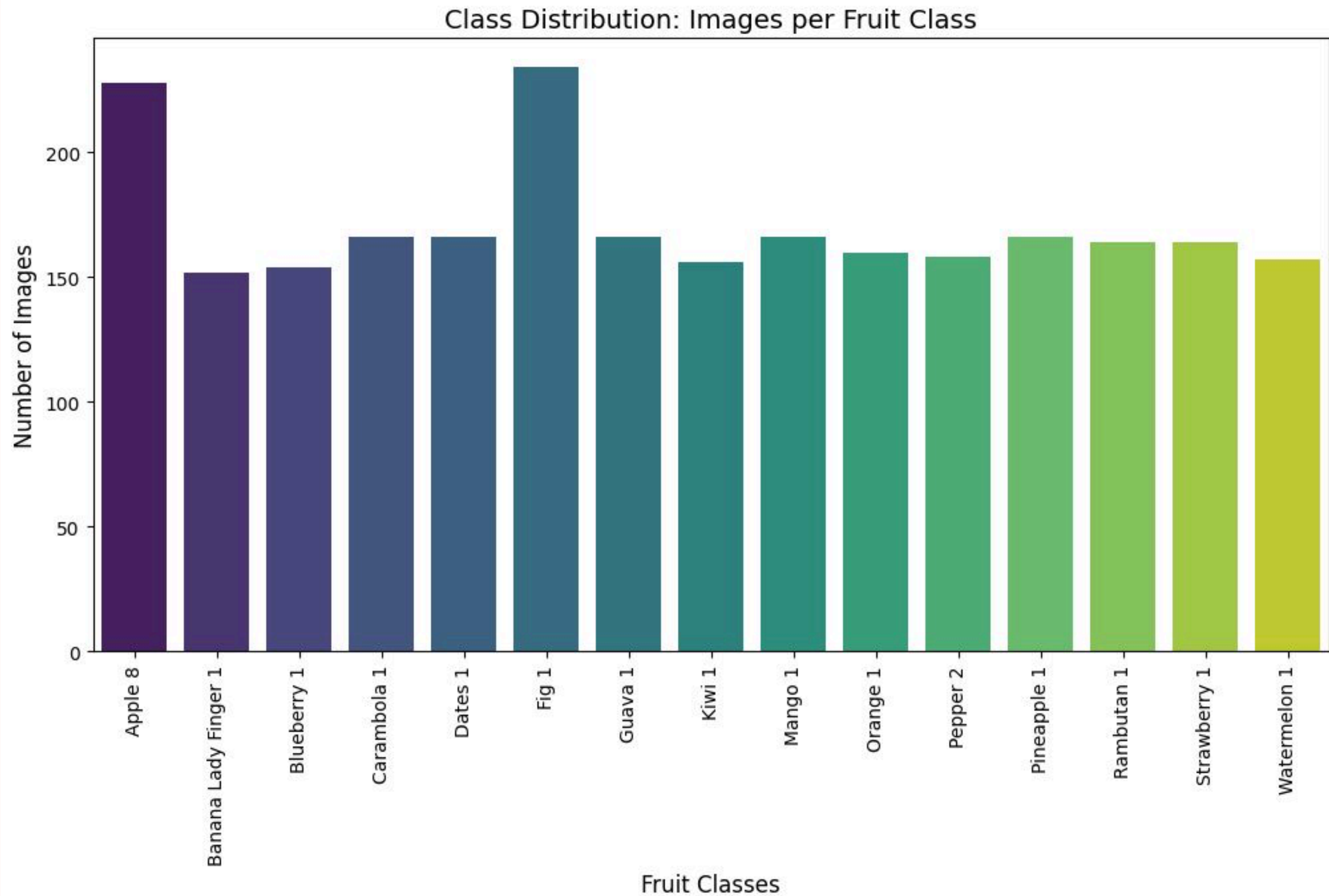
Image Format

All images are 100*100 pixels in RGB color, provided in JPG format

TRAIN DISTRIBUTION



TEST DISTRIBUTION



DATA AUGMENTATION

Balancing the dataset using Gaussian noise and blurring

